✅ Step 1 — Create the NEW ontology structure (from scratch or clean version)

👉 Either:

- create a **new ontology file**, OR
- reuse your previous one but **rename / clean classes**

🔹 Create the TOP structure

Go to **Classes tab → owl:Thing**

Add these top-level classes:

- NeurodegenerativeDisease
- Symptom
- SymptomCategory

🔹 Add diseases

Click NeurodegenerativeDisease → add:

- AlzheimerDisease
- ALS
- ParkinsonDisease

🔹 Add symptom categories

Click SymptomCategory → add:

- MotorSymptom
- CognitiveSymptom
- SpeechSwallowingSymptom
- BehaviouralPsychiatricSymptom
- AutonomicSymptom
- SleepSymptom
- SensorySymptom

◆ **Add core symptoms (IMPORTANT — keep it clean)**

Click Symptom → add ONLY these:

Motor (important for differentiation)

- Tremor
- RestingTremor
- Bradykinesia
- Rigidity
- GaitDisturbance
- PosturalInstability
- LimbWeakness
- MuscleWeakness

Speech / swallowing

- Dysarthria
- Dysphagia

Cognitive (Alzheimer core)

- MemoryImpairment
- EpisodicMemoryImpairment
- ImpairedReasoning
- LanguageImpairment

Behavioural / psychiatric

- Confusion
- Depression
- Anxiety
- Hallucination

Autonomic / other

- Constipation

Sleep

- SleepDisturbance

🧠 Why this is PERFECT

You just created:

- ✓ Clear
- ✓ Minimal
- ✓ Clinically meaningful
- ✓ Easy for chatbot

Instead of 100+ symptoms → you now have ~20 strong ones.

⛔ Stop here

Before we continue, confirm:

1. Do you have:
    - 3 diseases
    - 7 symptom categories
    - ~20 symptoms
2. Everything under the correct parent?

Next step

👉 Step 2 — Create **relationships (hasSymptom, moreTypicalOf, overlapsWith)**

This is where your chatbot becomes powerful 🔥

Step 2 — Create the relationships

Go to **Object Properties** and create these under owl:topObjectProperty:

- hasSymptom
- hasPrimarySymptom
- hasOverlappingSymptom
- moreTypicalOf
- belongsToCategory

We only really need the first four, but belongsToCategory is useful later if you add example patient cases or test cases.

Assign domain and range

1. hasSymptom

- Domain: NeurodegenerativeDisease
- Range: Symptom

Meaning: a disease can have a symptom.

2. hasPrimarySymptom

- Domain: NeurodegenerativeDisease
- Range: Symptom

Meaning: a disease has a core or especially characteristic symptom.

3. hasOverlappingSymptom

- Domain: NeurodegenerativeDisease
- Range: Symptom

Meaning: a disease has a symptom that can also appear in another disease.

4. moreTypicalOf

- Domain: Symptom
- Range: NeurodegenerativeDisease

Meaning: this symptom is more characteristic of one disease than the others.

5. belongsToCategory

- Domain: Symptom
- Range: SymptomCategory

Meaning: a symptom belongs to one symptom family, like MotorSymptom or CognitiveSymptom.

Why this matters

With these properties, your ontology can express things like:

- ParkinsonDisease hasPrimarySymptom RestingTremor
- ALS hasPrimarySymptom LimbWeakness
- AlzheimerDisease hasPrimarySymptom EpisodicMemoryImpairment
- Dysphagia moreTypicalOf ALS
- Tremor belongsToCategory MotorSymptom

That is exactly the kind of knowledge your chatbot needs.

What to do now in Protege

1. Go to Object Properties.
2. Add the 5 properties.
3. Set domain and range for each one.

Do only that for now.

When that is done, tell me and I'll give you **Step 3 — connect the diseases to the right symptoms**.

We are at the stage where you already created:

- classes
- object properties

Now we want to add:

1. disease individuals
2. symptom individuals
3. links between them

I'll break this into very small actions.

Step 3A — Create the disease individuals

What you are doing

You are creating one example instance for each disease:

- Alzheimer's
- ALS
- Parkinson's

These will be the "disease nodes" your chatbot knowledge can point to.

In Protege

1. Click the **Individuals** tab.
2. In the class hierarchy on the left, click **AlzheimerDisease**.
3. Click **Add individual**.
4. Name it:
    - alzheimers_case

Do the same for:

- under ALS → create als_case
- under ParkinsonDisease → create parkinsons_case

What you should have now

Under each disease class, you should see one individual:

- AlzheimerDisease
  - alzheimers_case
- ALS
  - als_case
- ParkinsonDisease
  - parkinsons_case

Stop here and make sure these three exist before going further.

Step 3B — Create the symptom individuals

What you are doing

For each symptom class, you will create one individual.
This makes the symptoms usable in relationships like:

- Parkinson hasPrimarySymptom resting tremor
- ALS hasPrimarySymptom limb weakness

Important note

You do **not** need to create individuals for every class in one go if that feels too long.
But I'll give you the exact list to enter.

In Protege

Go to the **Individuals** tab and do this class by class.

1. Under Tremor

Create:

- tremor_symptom

2. Under RestingTremor

Create:

- resting_tremor_symptom

3. Under Bradykinesia

Create:

- bradykinesia_symptom

4. Under Rigidity

Create:

- rigidity_symptom

5. Under GaitDisturbance

Create:

- gait_disturbance_symptom

6. Under PosturalInstability

Create:

- postural_instability_symptom

7. Under LimbWeakness

Create:

- limb_weakness_symptom

8. Under MuscleWeakness

Create:

- muscle_weakness_symptom

9. Under Dysarthria

Create:

- dysarthria_symptom

10. Under Dysphagia

Create:

- dysphagia_symptom

11. Under MemoryImpairment

Create:

- memory_impairment_symptom

12. Under EpisodicMemoryImpairment

Create:

- episodic_memory_impairment_symptom

13. Under ImpairedReasoning

Create:

- impaired_reasoning_symptom

14. Under LanguageImpairment

Create:

- language_impairment_symptom

15. Under Confusion

Create:

- confusion_symptom

16. Under Depression

Create:

- depression_symptom

17. Under Anxiety

Create:

- anxiety_symptom

18. Under Hallucination

Create:

- hallucination_symptom

19. Under Constipation

Create:

- constipation_symptom

20. Under SleepDisturbance

Create:

- sleep_disturbance_symptom

What you should have now

About 20 symptom individuals.

If this feels too long, do them in two batches:

- motor + speech first

- cognitive + behavioural + other second

Step 3C — Add symptom category links (belongsToCategory)

What you are doing

Now you connect each symptom individual to its symptom family.

Example:

- resting tremor belongs to MotorSymptom
- memory impairment belongs to CognitiveSymptom

In Protege

For each symptom individual:

1. Click the individual
2. Go to **Object property assertions**
3. Add:
    - property: belongsToCategory
    - target: the appropriate category individual/class target

Important clarification

If you did **not** create individuals for the symptom categories yet, do this first:

Under each symptom category class, create one individual:

- motor_category
- cognitive_category
- speech_swallowing_category
- behavioural_psychiatric_category
- autonomic_category
- sleep_category

Then use those as targets.

Add these links

Motor

These should point to motor_category:

- tremor_symptom

- resting_tremor_symptom
- bradykinesia_symptom
- rigidity_symptom
- gait_disturbance_symptom
- postural_instability_symptom
- limb_weakness_symptom
- muscle_weakness_symptom

Speech / swallowing

Point to speech_swallowing_category:

- dysarthria_symptom
- dysphagia_symptom

Cognitive

Point to cognitive_category:

- memory_impairment_symptom
- episodic_memory_impairment_symptom
- impaired_reasoning_symptom
- language_impairment_symptom
- confusion_symptom

Behavioural / psychiatric

Point to behavioural_psychiatric_category:

- depression_symptom
- anxiety_symptom
- hallucination_symptom

Autonomic

Point to autonomic_category:

- constipation_symptom

Sleep

Point to sleep_category:

- sleep_disturbance_symptom

Step 3D — Connect diseases to symptoms

This is the part that makes the ontology clinically useful.

In Protege

Click the disease individual, then add **Object property assertions**.

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1. For alzheimers_case

Add hasPrimarySymptom links to:

- episodic_memory_impairment_symptom
- memory_impairment_symptom
- impaired_reasoning_symptom
- language_impairment_symptom
- confusion_symptom

Add hasSymptom links to:

- depression_symptom
- anxiety_symptom
- sleep_disturbance_symptom

Add hasOverlappingSymptom link to:

- dysphagia_symptom

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2. For als_case

Add hasPrimarySymptom links to:

- limb_weakness_symptom
- muscle_weakness_symptom
- dysarthria_symptom
- dysphagia_symptom

Add hasSymptom link to:

- sleep_disturbance_symptom

Add hasOverlappingSymptom link to:

- confusion_symptom

---

3. For parkinsons_case

Add hasPrimarySymptom links to:

- resting_tremor_symptom
- bradykinesia_symptom
- rigidity_symptom
- gait_disturbance_symptom
- postural_instability_symptom

Add hasSymptom links to:

- dysarthria_symptom
- dysphagia_symptom
- constipation_symptom
- sleep_disturbance_symptom
- hallucination_symptom
- depression_symptom
- anxiety_symptom

Add hasOverlappingSymptom link to:

- confusion_symptom

Step 3E — Add moreTypicalOf links from symptom to disease

This lets the ontology answer questions like:

- Which disease is tremor more typical of?

In Protege

Click each symptom individual and add **Object property assertions** with property moreTypicalOf.

Add these:

Parkinson's-typical

- resting_tremor_symptom → parkinsons_case
- bradykinesia_symptom → parkinsons_case

- rigidity_symptom → parkinsons_case

ALS-typical

- limb_weakness_symptom → als_case
- muscle_weakness_symptom → als_case
- dysarthria_symptom → als_case

Alzheimer's-typical

- episodic_memory_impairment_symptom → alzheimers_case
- memory_impairment_symptom → alzheimers_case
- impaired_reasoning_symptom → alzheimers_case
- language_impairment_symptom → alzheimers_case

Step 3F — Quick check before moving on

At this point, your ontology should be able to represent things like:

- Parkinson's has primary symptom resting tremor
- ALS has primary symptom limb weakness
- Alzheimer's has primary symptom episodic memory impairment
- Dysphagia is an overlapping symptom
- Tremor belongs to the motor symptom category
- Resting tremor is more typical of Parkinson's

What I want you to do now

Complete these steps and then confirm:

1. Disease individuals created
2. Symptom individuals created
3. Category individuals created
4. Disease-to-symptom links added
5. Symptom-to-disease "moreTypicalOf" links added

Once you say it's done, I'll take you to the **next major step: building the 6 competency questions and mapping them to Dialogflow intents for this new doctor-support chatbot**.

We are at the stage where you already created:

- classes
- object properties

Now we want to add:

1. disease individuals
2. symptom individuals
3. links between them

I'll break this into very small actions.

Step 3A — Create the disease individuals

What you are doing

You are creating one example instance for each disease:

- Alzheimer's
- ALS
- Parkinson's

These will be the "disease nodes" your chatbot knowledge can point to.

In Protege

1. Click the **Individuals** tab.
2. In the class hierarchy on the left, click **AlzheimerDisease**.
3. Click **Add individual**.
4. Name it:
    - alzheimers_case

Do the same for:

- under ALS → create als_case
- under ParkinsonDisease → create parkinsons_case

What you should have now

Under each disease class, you should see one individual:

- AlzheimerDisease
    - alzheimers_case

- ALS
    - als_case
- ParkinsonDisease
    - parkinsons_case

Stop here and make sure these three exist before going further.

Step 3B — Create the symptom individuals

What you are doing

For each symptom class, you will create one individual.
This makes the symptoms usable in relationships like:

- Parkinson hasPrimarySymptom resting tremor
- ALS hasPrimarySymptom limb weakness

Important note

You do **not** need to create individuals for every class in one go if that feels too long.
But I'll give you the exact list to enter.

In Protege

Go to the **Individuals** tab and do this class by class.

1. Under Tremor

Create:

- tremor_symptom

2. Under RestingTremor

Create:

- resting_tremor_symptom

3. Under Bradykinesia

Create:

- bradykinesia_symptom

4. Under Rigidity

Create:

- rigidity_symptom

5. Under GaitDisturbance

Create:

- gait_disturbance_symptom

6. Under PosturalInstability

Create:

- postural_instability_symptom

7. Under LimbWeakness

Create:

- limb_weakness_symptom

8. Under MuscleWeakness

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- muscle_weakness_symptom

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- hallucination_symptom

19. Under Constipation

Create:

- constipation_symptom

20. Under SleepDisturbance

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- sleep_disturbance_symptom

What you should have now

About 20 symptom individuals.

If this feels too long, do them in two batches:

- motor + speech first
- cognitive + behavioural + other second

Step 3C — Add symptom category links (belongsToCategory)

What you are doing

Now you connect each symptom individual to its symptom family.

Example:

- resting tremor belongs to MotorSymptom
- memory impairment belongs to CognitiveSymptom

In Protege

For each symptom individual:

1. Click the individual
2. Go to **Object property assertions**
3. Add:
    - property: belongsToCategory
    - target: the appropriate category individual/class target

Important clarification

If you did **not** create individuals for the symptom categories yet, do this first:

Under each symptom category class, create one individual:

- motor_category
- cognitive_category
- speech_swallowing_category
- behavioural_psychiatric_category
- autonomic_category
- sleep_category

Then use those as targets.

Add these links

Motor

These should point to motor_category:

- tremor_symptom
- resting_tremor_symptom
- bradykinesia_symptom
- rigidity_symptom
- gait_disturbance_symptom
- postural_instability_symptom
- limb_weakness_symptom

- muscle_weakness_symptom

Speech / swallowing

Point to speech_swallowing_category:

- dysarthria_symptom
- dysphagia_symptom

Cognitive

Point to cognitive_category:

- memory_impairment_symptom
- episodic_memory_impairment_symptom
- impaired_reasoning_symptom
- language_impairment_symptom
- confusion_symptom

Behavioural / psychiatric

Point to behavioural_psychiatric_category:

- depression_symptom
- anxiety_symptom
- hallucination_symptom

Autonomic

Point to autonomic_category:

- constipation_symptom

Sleep

Point to sleep_category:

- sleep_disturbance_symptom

Step 3D — Connect diseases to symptoms

This is the part that makes the ontology clinically useful.

In Protege

Click the disease individual, then add **Object property assertions**.

---

1. For alzheimers_case

Add hasPrimarySymptom links to:

- episodic_memory_impairment_symptom
- memory_impairment_symptom
- impaired_reasoning_symptom
- language_impairment_symptom
- confusion_symptom

Add hasSymptom links to:

- depression_symptom
- anxiety_symptom
- sleep_disturbance_symptom

Add hasOverlappingSymptom link to:

- dysphagia_symptom

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2. For als_case

Add hasPrimarySymptom links to:

- limb_weakness_symptom
- muscle_weakness_symptom
- dysarthria_symptom
- dysphagia_symptom

Add hasSymptom link to:

- sleep_disturbance_symptom

Add hasOverlappingSymptom link to:

- confusion_symptom

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3. For parkinsons_case

Add hasPrimarySymptom links to:

- resting_tremor_symptom
- bradykinesia_symptom

- rigidity_symptom
- gait_disturbance_symptom
- postural_instability_symptom

Add hasSymptom links to:

- dysarthria_symptom
- dysphagia_symptom
- constipation_symptom
- sleep_disturbance_symptom
- hallucination_symptom
- depression_symptom
- anxiety_symptom

Add hasOverlappingSymptom link to:

- confusion_symptom

Step 3E — Add moreTypicalOf links from symptom to disease

This lets the ontology answer questions like:

- Which disease is tremor more typical of?

In Protege

Click each symptom individual and add **Object property assertions** with property moreTypicalOf.

Add these:

Parkinson's-typical

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- bradykinesia_symptom → parkinsons_case
- rigidity_symptom → parkinsons_case

ALS-typical

- limb_weakness_symptom → als_case
- muscle_weakness_symptom → als_case
- dysarthria_symptom → als_case

Alzheimer's-typical

- episodic_memory_impairment_symptom → alzheimers_case
- memory_impairment_symptom → alzheimers_case
- impaired_reasoning_symptom → alzheimers_case
- language_impairment_symptom → alzheimers_case

Step 3F — Quick check before moving on

At this point, your ontology should be able to represent things like:

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- ALS has primary symptom limb weakness
- Alzheimer's has primary symptom episodic memory impairment
- Dysphagia is an overlapping symptom
- Tremor belongs to the motor symptom category
- Resting tremor is more typical of Parkinson's

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Complete these steps and then confirm:

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